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When Do Graph Neural Networks Help?

Class 5 — First Research Results

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01 Project reminder

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Research question

When do GNNs beat MLP / XGBoost, and how much edge noise can they survive before losing?

- **H1:** on clean similar graphs GNNs win; after ∼30% fake edges, tabular models catch up.
- **H2:** on mixed graphs, tabular models may win from the start.

Class 5 focus: first hypothesis test (Exp. A) + partial Exp. B + what failed / surprised us + next controls.

02 What we bring today



Done (reproducible)

- Data audit (Cora)
- Clean baseline (5 seeds)
- Exp. A edge deletion (5 seeds)
- Exp. B fake edges — *partial* (3 seeds)

Fair protocol

- Same Planetoid masks
- MLP / XGBoost / GCN
- Accuracy + macro-F1
- Saved under `RESULTS/`

03 Clean Cora baseline (reminder)

Test accuracy — mean ± std over 5 seeds

Model	Test accuracy	Test macro-F1
XGBoost	0.527 ± 0.006	0.514 ± 0.006
MLP	0.591 ± 0.008	0.572 ± 0.008
GCN	0.818 ± 0.005	0.806 ± 0.005

Starting point of **H1**: with good (similar) connections, the graph helps ($\approx +23$ pp vs MLP).

04 Experiment A — delete real edges



Protocol: randomly remove undirected edges $0\%, 20\%, 40\%, 60\%, 80\%, 100\%$ (5 seeds).

- Tabular models ignore edges $\Rightarrow$ should stay flat
- GCN should fall as structure disappears
- At 100% deletion GCN $\approx$ feature-only model

05 Experiment A results



Test accuracy (mean, 5 seeds)

Model	0%	20%	40%	60%	80%	100%
XGBoost	0.527	0.527	0.527	0.527	0.527	0.527
MLP	0.593	0.592	0.592	0.592	0.592	0.592
GCN	**0.818**	0.793	0.764	0.726	0.673	0.594

`RESULTS/expA_cora_summary.csv`

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06 Experiment A — takeaways

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- MLP / XGBoost stay **flat** (control works).
- GCN drops smoothly: 0.82 $\rightarrow$ 0.59.
- Even after **80%** edge removal, GCN still beats MLP (0.67 $>$ 0.59).
- At **100%**, GCN $\approx$ MLP (0.594 vs 0.592).

Hypothesis status

Removing real structure hurts GCN as expected, but Cora stays useful for GCN until almost all edges are gone.

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07 Experiment B — add fake edges



Protocol: keep all real edges; add random non-edges $+0\%, 15\%, 30\%, 45\%, 60\%, 75\%$ of the real edge count.

- **H1 claim:** after $\sim$30% fake edges, tabular overtakes GCN
- Status today: **partial** — 3 seeds (0–2), same models
- Code: `scripts/run_experiment_b.py`

08 Experiment B — partial results (Cora)

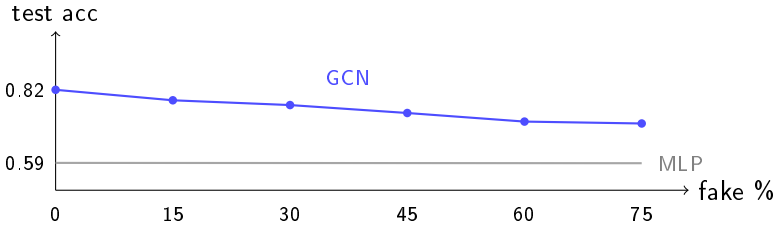


Test accuracy (mean over 3 seeds)

Model	0%	15%	30%	45%	60%	75%
XGBoost	0.529	0.529	0.529	0.529	0.529	0.529
MLP	0.586	0.585	0.585	0.585	0.585	0.585
GCN	**0.816**	0.783	0.768	0.743	0.716	0.710

`RESULTS/expB_cora_summary.csv` — mark as partial (n=3).

09 Experiment B — sketch



GCN falls, but stays clearly above MLP through +**75%** fake edges.

10 What surprised us (“failures”)

H1 breaking point ∼30% — not confirmed on Cora

At +30% fake edges GCN is still ≈ 0.77 vs MLP ≈ 0.59. Even at +75% GCN ≈ 0.71 — tabular does *not* overtake.

- Cora is strongly similar (∼81% same-class neighbours) ⇒ real edges still dominate under this noise protocol.
- Exp. A already showed the same pattern: advantage shrinks slowly.
- Negative / surprising result is useful — we revise the claim, not the protocol.

11 Updated reading of H1



1. On clean Cora, GNN clearly helps — **supported**.
2. Edge deletion / fake edges degrade GCN — **supported**.
3. "∼30% fake edges ⇒ tabular wins" — **too aggressive for Cora**; revise threshold or noise type.

What would change our mind next

If GCN still wins on a **mixed** graph (Roman-empire / filtered Chameleon), then H2 is also weakened.

12 Next controlled experiments



1. Finish Exp. B: extend to **5 seeds**; optional higher fake fractions (e.g. 100–200%) or denser noise schedule.
2. **Mixed dataset** (H2): Roman-empire or filtered Chameleon.
3. Add **GraphSAGE** (planned main GNN for Exp. B); optional GAT.
4. Class 6: ablations, uncertainty, draft of final deck.

13 Class 5 takeaway

- First hypothesis test (Exp. A) is complete and reproducible.
- Partial Exp. B already shows GCN is **robust** to substantial fake-edge noise on Cora.
- We treat the missed $\sim$30% breaking point as a finding, not a failed project.

One sentence

On Cora the graph helps a lot and survives a lot of noise; the interesting failure mode may live on mixed graphs, not on mild fake-edge corruption.